

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

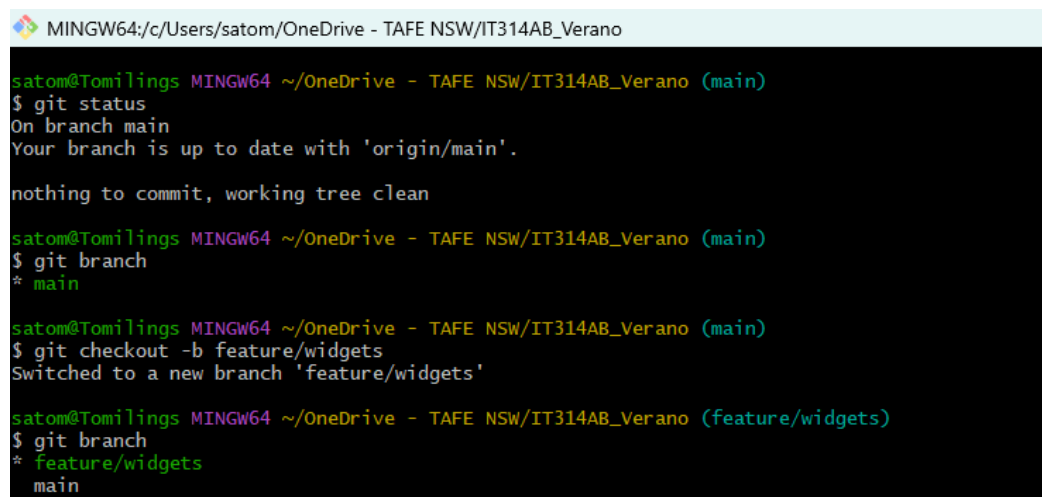
(I know what AI-generated code looks like, don't even think about it :>)

Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`



```
MINGW64:/c/Users/satom/OneDrive - TAFE NSW/IT314AB_Verano
satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (main)
$ git branch
* main

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (main)
$ git checkout -b feature/widgets
Switched to a new branch 'feature/widgets'

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/widgets)
$ git branch
* feature/widgets
  main
```

Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

```
Text(  
  'Chiara Marie C. Canque\n'  
  'BSIT - 3\n'  
  'My First Flutter Application\n'  
  'August 11, 2026',  
)
```

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

`Text` → Name

`Text` → Course & Section

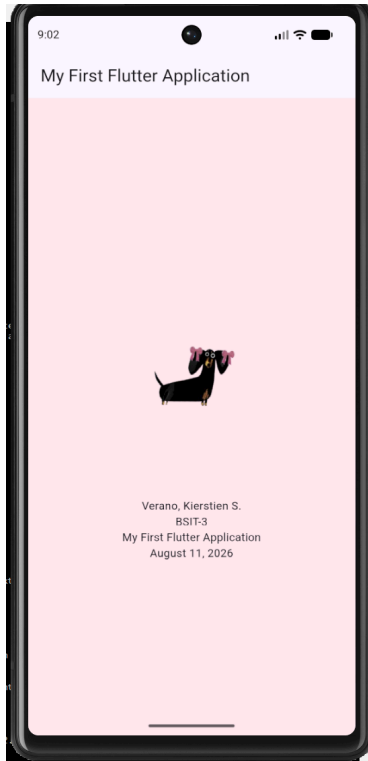
`Text` → Application Title

`Text` → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application



```
29 Text("Verano, Kierstien S."),  
30 Text("BSIT-3"),  
31 Text("My First Flutter Application"),  
32 Text("August 11, 2026"),
```

🚩 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

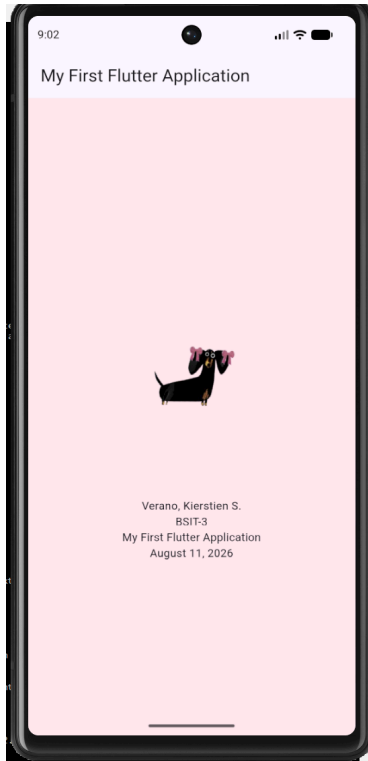
Course & Section

Application Title

Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically



```
20 child: Column(  
21   mainAxisAlignment: MainAxisAlignment.center,  
22   children: [  
23     Image.asset(  
24       'assets/dachshund.png',  
25       width: 150,  
26     ), // Image.asset  
27     SizedBox(height: 20),  
28  
29     Text("Verano, Kierstien S."),  
30     Text("BSIT-3"),  
31     Text("My First Flutter Application"),  
32     Text("August 11, 2026"),  
33   ],  
34 ), // Column
```

🚩 Flag 3 – Add Another Element

Objective: Add another individual widget to your `Column`.

Add something that belongs to your profile.

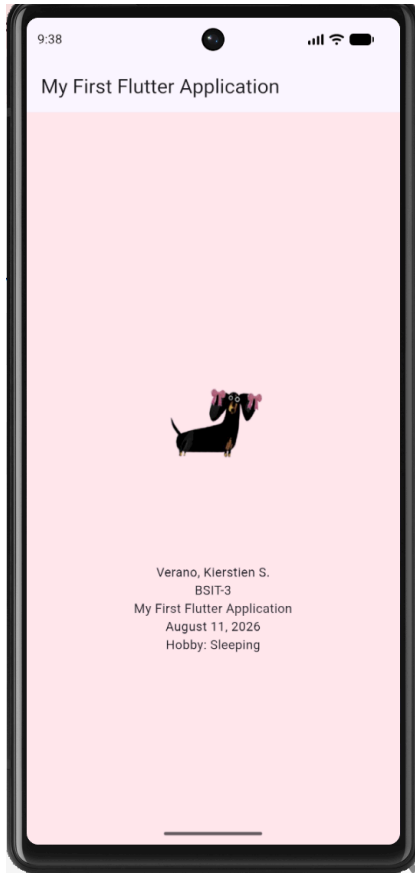
Examples:

Hobby

It must be its **own widget**, not additional text inside another `Text`.

Screenshot:

- New widget in your `Column`
- Updated application



```
20 child: Column(  
21   mainAxisAlignment: MainAxisAlignment.center,  
22   children: [  
23     Image.asset(  
24       'assets/dachshund.png',  
25       width: 150,  
26     ), // Image.asset  
27     SizedBox(height: 20),  
28  
29     Text("Verano, Kierstien S."),  
30     Text("BSIT-3"),  
31     Text("My First Flutter Application"),  
32     Text("August 11, 2026"),  
33     Text("Hobby: Sleeping"),  
34   ],
```

🚩 Flag 4 – Put Two Things Side-by-Side

Objective: Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

`Column`

Text

Text

Text

Text

Text

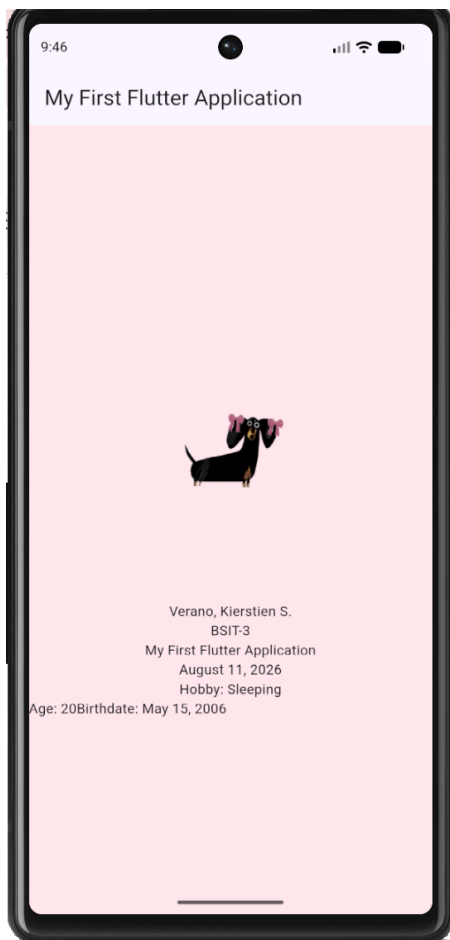
Row

Text

Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side



```
35 Row(  
36   children: [  
37     Text('Age: 20'),  
38     Text('Birthdate: May 15, 2006'),  
39   ],  
40 ), // Row
```

🚩 Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizeBox()`

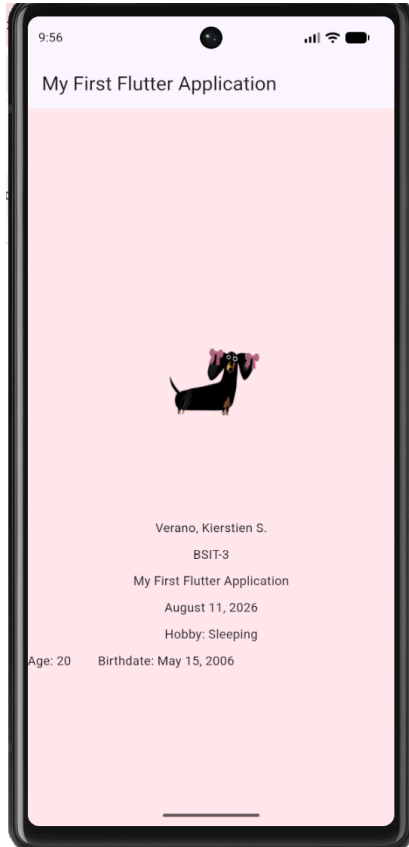
Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet.**

Screenshot:

- `SizeBox` usage
- Updated application



```
20 child: Column(  
21   mainAxisAlignment: MainAxisAlignment.center,  
22   children: [  
23     Image.asset(  
24       'assets/dachshund.png',  
25       width: 150,  
26     ), // Image.asset  
27     SizeBox(height: 20),  
28  
29     Text("Verano, Kierstien S."),  
30     SizeBox(height: 10),  
31     Text("BSIT-3"),  
32     SizeBox(height: 10),  
33     Text("My First Flutter Application"),  
34     SizeBox(height: 10),  
35     Text("August 11, 2026"),  
36     SizeBox(height: 10),  
37     Text("Hobby: Sleeping"),  
38     SizeBox(height: 10),  
39  
40     Row(  
41       children: [  
42         Text('Age: 20'),  
43         SizeBox(width: 30),  
44         Text('Birthdate: May 15, 2006'),  
45       ],  
46     ), // Row  
47   ],  
48 ), // Column
```

🚩 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

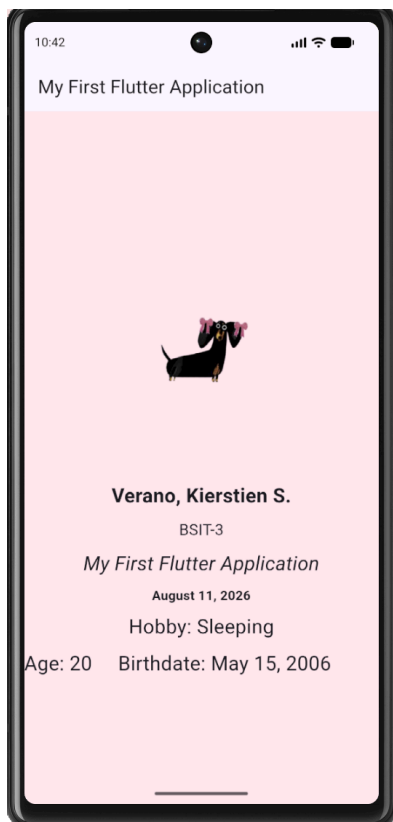
Apply different styling to at least **three** elements.

For example:

- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes
- Updated application



```
29 Text(  
30   "Verano, Kierstien S.",  
31   style: TextStyle(  
32     fontSize: 23,  
33     fontWeight: FontWeight.bold,  
34   ), // TextStyle  
35 ), // Text  
36 SizedBox(height: 10),  
37  
38 Text(  
39   "BSIT-3",  
40   style: TextStyle(  
41     fontSize: 18,  
42   ), // TextStyle  
43 ), // Text  
44 SizedBox(height: 10),  
45  
46 Text(  
47   "My First Flutter Application",  
48   style: TextStyle(  
49     fontSize: 23,  
50     fontStyle: FontStyle.italic,  
51   ), // TextStyle  
52 ), // Text  
53  
54 SizedBox(height: 10),  
55 Text(  
56   "August 11, 2026",  
57   style: TextStyle(  
58     fontSize: 15,  
59     fontWeight: FontWeight.bold,  
60   ), // TextStyle  
61 ), // Text  
62 SizedBox(height: 10),  
63  
64 Text(  
65   "Hobby: Sleeping",  
66   style: TextStyle(  
67     fontSize: 23,  
68   ), // TextStyle  
69 ), // Text  
70 SizedBox(height: 10),  
71
```

```
72 Row(  
73   children: [  
74     Text(  
75       'Age: 20',  
76       style: TextStyle(  
77         fontSize: 23,  
78       ), // TextStyle  
79     ), // Text  
80     SizedBox(width: 30),  
81     Text(  
82       'Birthdate: May 15, 2006',  
83       style: TextStyle(  
84         fontSize: 23,  
85       ), // TextStyle  
86     ), // Text  
87   ],  
88 ), // Row  
89 ],  
90 ), // Column
```


🚩 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

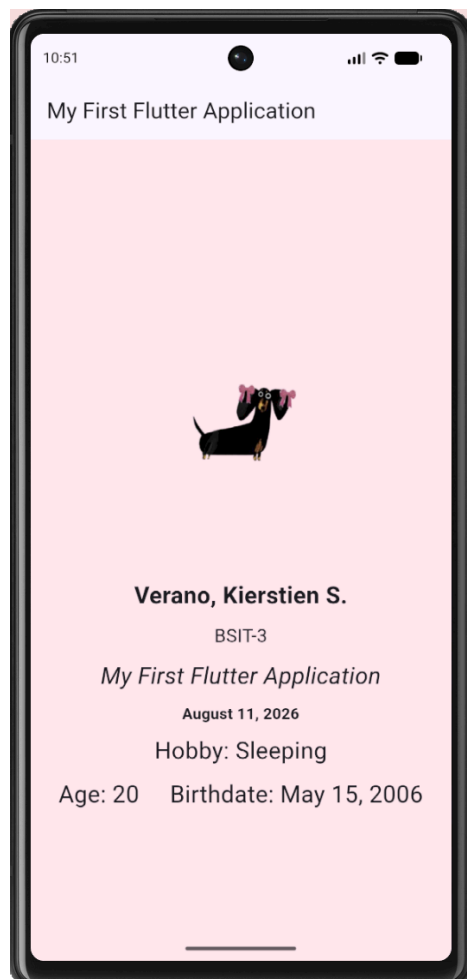
`mainAxisAlignment`

`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed
- Updated application



```
72  
73  
74  
75  
76  
77  
78  
79  
80  
81  
82  
83  
84  
85  
86  
87  
88  
89  
90  
Row(  
  mainAxisAlignment: MainAxisAlignment.center,  
  crossAxisAlignment: CrossAxisAlignment.center,  
  children: [  
    Text(  
      'Age: 20',  
      style: TextStyle(  
        fontSize: 23,  
      ), // TextStyle  
    ), // Text  
    SizedBox(width: 30),  
    Text(  
      'Birthdate: May 15, 2006',  
      style: TextStyle(  
        fontSize: 23,  
      ), // TextStyle  
    ), // Text  
  ],  
) // Row
```

Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second **Card** containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

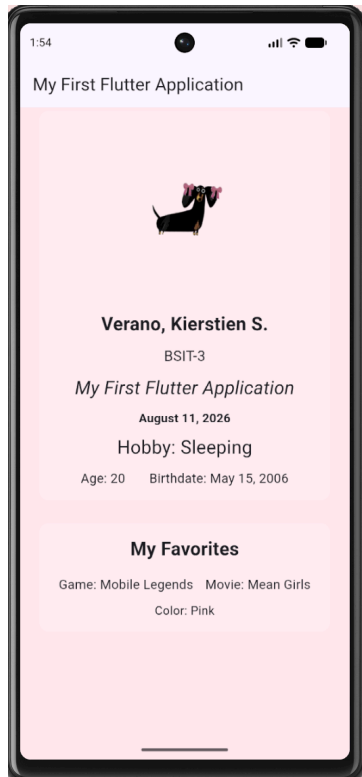
- Individual **Text** widgets
- A **Column**
- At least one **Row**
- Spacing

Screenshot:

- Code for the second card
- Both cards displayed

```
112 // Card 2: Favorites of mine <33
113 Card(
114   color: Colors.white.withOpacity(0.3),
115   elevation: 0,
116   child: Padding(
117     padding: const EdgeInsets.all(16),
118     child: Column(
119       mainAxisAlignment: MainAxisAlignment.center,
120       children: [
121         Text(
122           "My Favorites",
123           style: TextStyle(
124             fontSize: 23,
125             fontWeight: FontWeight.bold,
126           ), // TextStyle
127         ), // Text
128         SizedBox(height: 16),
129         Row(
130           mainAxisAlignment: MainAxisAlignment.center,
131           crossAxisAlignment: CrossAxisAlignment.center,
132           children: [
133             Text(
134               'Game: Mobile Legends',
135               style: TextStyle(
136                 fontSize: 16,
137               ), // TextStyle
138             ), // Text
139             SizedBox(width: 15),
140             Text(
141               'Movie: Mean Girls',
142               style: TextStyle(
143                 fontSize: 16,
144               ), // TextStyle
145             ), // Text
146           ],
147         ), // Row
148       ],
149     ), // Card
150   ),
```

```
151     SizedBox(height: 10),
152     Text(
153       "Color: Pink",
154       style: TextStyle(
155         fontSize: 15,
156       ), // TextStyle
157     ), // Text
158   ], // Column
159 ), // Padding
160 ), // Card
161 ], // Scaffold
162 ), // SingleChildScrollView
163 ), // MaterialApp
164 ), // MaterialApp
165 ), // Scaffold
166 ), // SingleChildScrollView
167 ), // MaterialApp
168 ), // MaterialApp
169 ), // MaterialApp
170 ), // MaterialApp
171 ), // MaterialApp
172 ), // MaterialApp
```



🚩 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card



Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application

```

19 body: SingleChildScrollView( // provides scrolling capability to a single child container whose content might exceed the available screen space
20   child: Padding(
21     padding: const EdgeInsets.symmetric(horizontal: 20),
22     child: Column(
23       mainAxisAlignment: MainAxisAlignment.center,
24       crossAxisAlignment: CrossAxisAlignment.stretch,
25       children: [
26
27         // Card 1: Personal Information (Profile kemuru of mine)
28         Card(
29           color: Colors.white.withOpacity(0.3),
30           elevation: 0,
31           child: Padding(
32             padding: const EdgeInsets.fromLTRB(16, 8, 16, 16),
33             child: Column(
34               mainAxisAlignment: MainAxisAlignment.center,
35               children: [
36                 Image.asset(
37                   'assets/dachshund.png',
38                   width: 130,
39                 ), // Image.asset
40
41                 SizedBox(height: 10),
42                 Text(
43                   "Verano, Kierstien S.",
44                   style: TextStyle(
45                     fontSize: 23,
46                     fontWeight: FontWeight.bold,
47                   ), // TextStyle
48                 ), // Text
49
50                 SizedBox(height: 10),
51                 Text(
52                   "BSIT-3",
53                   style: TextStyle(
54                     fontSize: 18,
55                   ), // TextStyle
56                 ), // Text
57

```

```

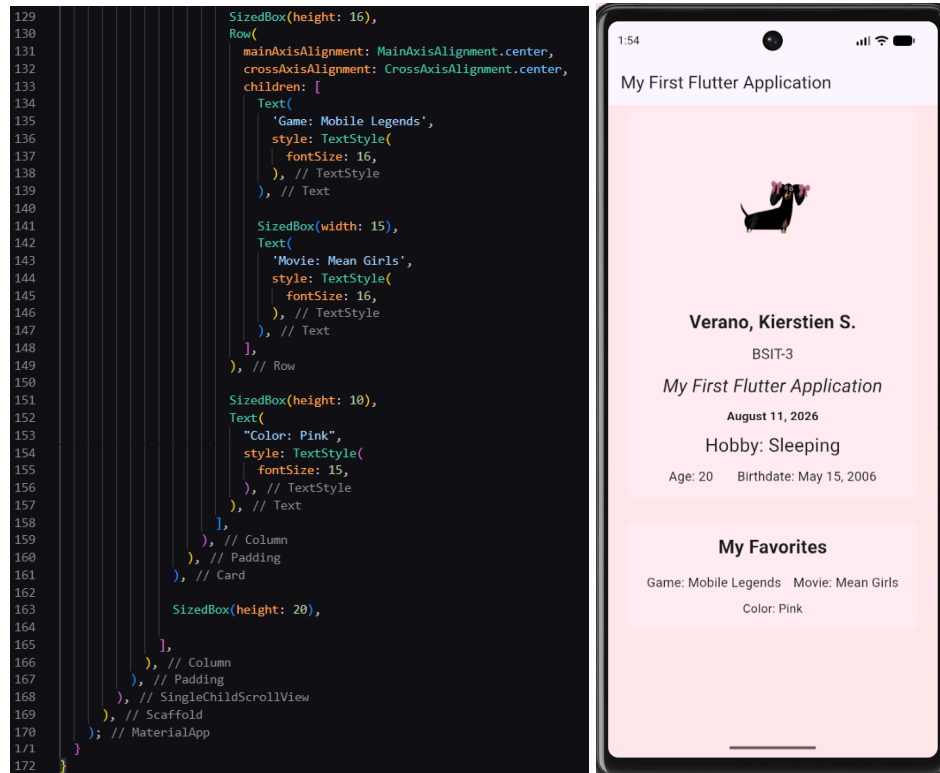
58                 SizedBox(height: 10),
59                 Text(
60                   "My First Flutter Application",
61                   style: TextStyle(
62                     fontSize: 23,
63                     fontStyle: FontStyle.italic,
64                   ), // TextStyle
65                 ), // Text
66
67                 SizedBox(height: 10),
68                 Text(
69                   "August 11, 2026",
70                   style: TextStyle(
71                     fontSize: 15,
72                     fontWeight: FontWeight.bold,
73                   ), // TextStyle
74                 ), // Text
75
76                 SizedBox(height: 10),
77                 Text(
78                   "Hobby: Sleeping",
79                   style: TextStyle(
80                     fontSize: 23,
81                   ), // TextStyle
82                 ), // Text
83
84                 SizedBox(height: 10),
85                 Row(
86                   mainAxisAlignment: MainAxisAlignment.center,
87                   crossAxisAlignment: CrossAxisAlignment.center,
88                   children: [
89                     Text(
90                       'Age: 20',
91                       style: TextStyle(
92                         fontSize: 16,
93                       ), // TextStyle
94                     ), // Text
95

```

```

96                 SizedBox(width: 30),
97                 Text(
98                   'Birthdate: May 15, 2006',
99                   style: TextStyle(
100                     fontSize: 16,
101                   ), // TextStyle
102                 ), // Text
103               ],
104             ), // Row
105           ],
106         ), // Column
107       ), // Padding
108     ), // Card
109
110     SizedBox(height: 20),
111
112     // Card 2: Favorites of mine <33
113     Card(
114       color: Colors.white.withOpacity(0.3),
115       elevation: 0,
116       child: Padding(
117         padding: const EdgeInsets.all(16),
118         child: Column(
119           mainAxisAlignment: MainAxisAlignment.center,
120           children: [
121             Text(
122               "My Favorites",
123               style: TextStyle(
124                 fontSize: 23,
125                 fontWeight: FontWeight.bold,
126               ), // TextStyle
127             ), // Text
128

```



🚩 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

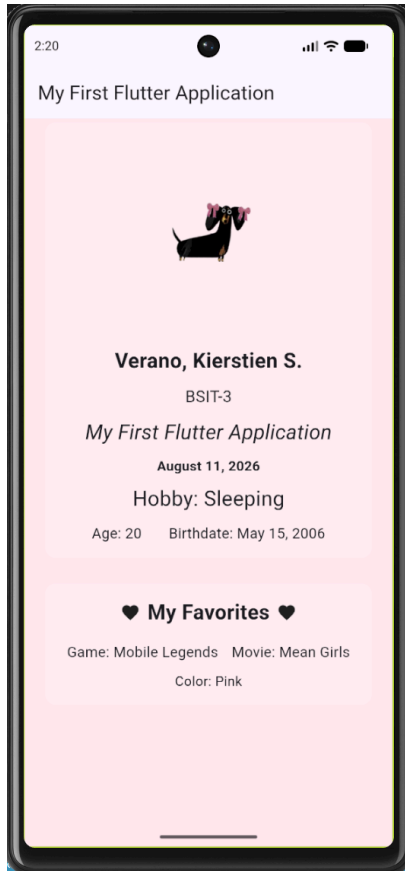
Examples:

- Icon
- Divider
- CircleAvatar
- Chip
- ElevatedButton

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application



```
102  
103  
104  
105  
106  
107  
108  
109  
110  
111  
112  
113  
114  
115  
116  
117  
children: [  
  Icon(Icons.favorite, size: 22),  
  
  SizedBox(width: 8),  
  Text(  
    "My Favorites",  
    style: TextStyle(  
      fontSize: 23,  
      fontWeight: FontWeight.bold,  
    ), // TextStyle  
  ), // Text  
  
  SizedBox(width: 8),  
  Icon(Icons.favorite, size: 22),  
],  
) // Row
```

🚩 Flag 11 – Real Phone Test 📱

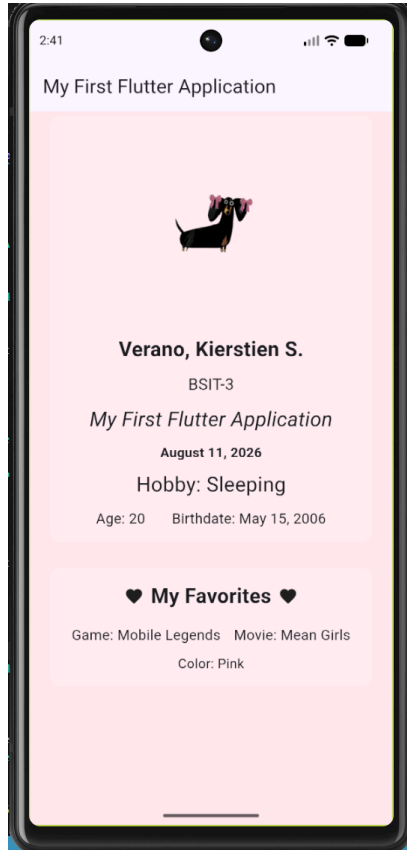
Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)

NOTE: CURRENT PHONE - IPHONE



🚩 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- `feature/widgets` branch

```

PS C:\Users\satom\OneDrive - TAFE NSW\IT314AB_Verano\FlutterProjects\my_first_flutter_app> git branch
* feature/widgets
  main
PS C:\Users\satom\OneDrive - TAFE NSW\IT314AB_Verano\FlutterProjects\my_first_flutter_app> git status
On branch feature/widgets
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   lib/main.dart

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\satom\OneDrive - TAFE NSW\IT314AB_Verano\FlutterProjects\my_first_flutter_app> git add .
PS C:\Users\satom\OneDrive - TAFE NSW\IT314AB_Verano\FlutterProjects\my_first_flutter_app> git status
On branch feature/widgets
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   lib/main.dart

PS C:\Users\satom\OneDrive - TAFE NSW\IT314AB_Verano\FlutterProjects\my_first_flutter_app> git commit -m "feat: add multiple cards and widgets"
[feature/widgets 28de1e3] feat: add multiple cards and widgets
 1 file changed, 133 insertions(+), 39 deletions(-)
PS C:\Users\satom\OneDrive - TAFE NSW\IT314AB_Verano\FlutterProjects\my_first_flutter_app> git status
On branch feature/widgets
nothing to commit, working tree clean

```

```

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano/FlutterProjects/my_first_flutter_app (feature/widgets)
● $ git branch
* feature/widgets
  main

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano/FlutterProjects/my_first_flutter_app (feature/widgets)
● $ git remote -v
origin  https://github.com/KVerano/IT314AB_Verano.git (fetch)
origin  https://github.com/KVerano/IT314AB_Verano.git (push)

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano/FlutterProjects/my_first_flutter_app (feature/widgets)
● $ git push -u origin feature/widgets
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 1.76 KiB | 1.76 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/widgets' on GitHub by visiting:
remote:   https://github.com/KVerano/IT314AB_Verano/pull/new/feature/widgets
remote:
To https://github.com/KVerano/IT314AB_Verano.git
 * [new branch]      feature/widgets -> feature/widgets
branch 'feature/widgets' set up to track 'origin/feature/widgets'.
🔔
satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano/FlutterProjects/my_first_flutter_app (feature/widgets)
○ $

```